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Self-Improvement of Language Models by Post-Training on Multi-Agent Debate

Ankur Samanta[1,2,*], **Akshayaa Magesh**[1], **Runzhe Wu**[1,3,*], **Ayush Jain**[1], **Youliang Yu**[1], **Daniel Jiang**[1], **Boris Vidolov**[1], **Paul Sajda**[2], **Yonathan Efroni**[1,†], **Kaveh Hassani**[4,†]

[1]Meta AI, [2]Columbia University, [3]Cornell Tech, [4]Meta Superintelligence Labs
[*]Work done at Meta, [†]Joint last author, alphabetical order

Self-improvement, where models improve beyond their current performance without external supervision, remains a challenge. The core difficulty is sourcing a training signal stronger than what the model itself can currently produce. Majority voting has been shown to provide such a signal by aggregating over multiple samples, helping mitigate some of the inconsistencies in LM reasoning. In this work, we show that multi-agent debate—where models collaborate and exchange reasoning over multiple rounds—provides an even richer signal than single-round majority voting. We introduce Multi-Agent Consensus Alignment (MACA), which uses reinforcement learning (RL) to post-train models to effectively utilize multi-agent debate. We find that preference learning over full reasoning traces, learning to differentiate between majority and minority reasoning, is more effective than binary consensus rewards or SFT-based approaches for leveraging these debate signals. This produces three key improvements: models are (1) better at utilizing the multi-agent debate setting (+26.87% on MATH), (2) individually more accurate (+21.51% on MathQA), and (3) more self-consistent (+27.6% on GSM8K). We also see strong generalization to unseen benchmarks (+16.3% on GPQA, +11.6% on CommonsenseQA).

Date: February 2, 2026
Correspondence: Ankur Samanta at as7416@columbia.edu
Code: https://github.com/facebookresearch/maca

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1 Introduction

Self-improvement in language models, where models train on self-generated signals without external supervision, is challenging. LMs suffer from inconsistent reasoning, often producing contradictory solutions to the same problem under exploratory sampling (Holtzman et al., 2020; Wang et al., 2023). While inference-time aggregation methods such as majority voting and multi-agent debate help improve the reliability of the final response (Wang et al., 2023; Li et al., 2024; Irving et al., 2018), they do not address the core problem: the internal reasoning instability that produces these inconsistencies in the first place. We argue that self-consistency, the ability to generate consistent outputs across diverse reasoning paths (Elazar et al., 2021; Wang et al., 2023), is a property of well-aligned reasoning systems. In the brain, consistency emerges through dedicated control mechanisms in prefrontal and anterior cingulate regions that resolve conflicts between competing neural activations, directing cognition toward coherent solutions (Miller and Cohen, 2001; Friston, 2010; Botvinick et al., 2004; Shenhav et al., 2013; Zhang et al., 2025). Language models, by contrast, lack mechanisms to align their diverse reasoning paths toward consensus.

Prior work shows that majority-vote signals can be an effective self-improvement signal (Wang et al., 2023; Zuo et al., 2025; Prasad et al., 2025). However, the models' underlying poor self-consistency can reduce the efficacy of such reinforcement. When models sample low-quality, inconsistent reasoning traces, aggregation can be counterproductive: noisy arguments compound rather than cancel out (Radharapu et al., 2025). We show that this problem is particularly acute in small language models, where simple statistical aggregation can compound low-quality reasoning and collaboration can result in weak consensus.

To achieve effective self-improvement in language models, we introduce Multi-Agent Consensus Alignment (MACA), a reinforcement learning framework that post-trains models using debate-derived signals to strengthen self-consistency and collaborative reasoning. Through multi-agent debate, models observe which reasoning patterns converge across peers and learn to favor those patterns through reinforcement on debate-derived consensus signals. This creates a

positive feedback loop: improved self-consistency and ability to utilize peer context enables more effective debate, which generates higher-quality training signals for the next iteration (Fig. 1).

A key design question in MACA is how to translate the multi-agent debate into a training signal. We explore the use of majority/minority labels as rewards through majority-vote GRPO (Shao et al., 2024), as well as using them as preference labels for DPO (Rafailov et al., 2024) and KTO (Ethayarajh et al., 2024). We find that preference learning is more effective: while both use majority/minority labels, preference learning over full reasoning traces allows models to learn to differentiate between reasoning traces that survived through deliberation and those that did not, capturing patterns where agents successfully reference and build on each other's reasoning. Crucially, this requires no ground-truth labels. Preference learning through relative comparison naturally grounds reasoning in the absence of external verification: when truth is ambiguous, judgments become robust through comparative assessment where majority reasoning provides a normative anchor while minority views introduce necessary variation (Festinger, 1954; Moscovici and Lage, 1976; Nemeth, 1986; Condorcet, 1972). We teach models to ground their reasoning through this comparison of debate-derived trajectories, learning stable reasoning through peer deliberation alone.

Key contributions.

- **Method.** We introduce MACA, a reinforcement learning framework that post-trains models on consensus signals derived from multi-agent debate, teaching them to internalize collaborative reasoning patterns.
- **Insight.** We demonstrate that consensus signals from multi-agent debate, grounded in peer interaction, provide richer and more robust training signals for self-improvement than single-round majority voting. This creates a positive feedback loop where improved reasoning enables more effective debate, which generates higher-quality training signals.
- **Empirical validation.** We show that MACA improves multi-agent collaborative performance (+26.87% on MATH), individual reasoning (+21.51% on MathQA), and self-consistency (+27.6% on GSM8K), with strong generalization to unseen reasoning domains (+16.3% on GPQA, +11.6% on CommonsenseQA).

2 Related Work

Existing approaches address sampling inconsistency primarily through inference-time techniques. Self-consistency prompting (Wang et al., 2023; Li et al., 2024) samples multiple reasoning paths and selects the majority-voted answer, with extensions for non-verifiable outputs (Chen et al., 2023) and path pruning (Zhu et al., 2025a). Multi-agent debate frameworks (Du et al., 2023; Irving et al., 2018) similarly utilize consensus across models to improve reliability, with recent work exploring applications in scientific discovery (Gottweis et al., 2025). However, these methods require additional inference compute and do not internalize self-consistency into the model. To improve self-consistency and single rollout performance through post-training, current training-time alternatives have limitations: relative log-probability rankings as a proxy for relative confidence correlate less strongly with accuracy than majority vote signals, as we show (App. H), while LLM-as-a-Judge approaches (Jiao et al., 2025) suffer from preference leakage (Li et al., 2025) and bias under ambiguity (Radharapu et al., 2025). We improve self-consistency through post-training that optimizes consensus signals via multi-agent RL (Yang et al., 2021; Jiang and Lu, 2023; Zhu et al., 2025b; Zhan et al., 2024). Majority vote RL methods such as TTRL (Zuo et al., 2025) and ScPO (Prasad et al., 2025) use GRPO and DPO, respectively, to reinforce single-round majority vote from independently sampled responses. By contrast, MACA introduces multi-agent, multi-round debate, including deliberation traces as conditioning contexts within the prompts during RL. This exposes models to concrete examples of relative grounding where consensus emerges through explicit engagement with peer arguments, allowing them to learn not just to aggregate outputs but to internalize the deliberative process by which consensus is reached. Both TTRL and ScPO are specialized cases of the framework developed in this work, when reducing the multi-agent debate parameters to a single-round majority vote. For post-training the multi-agent debate setting specifically, Subramaniam et al. (2025) uses Supervised Fine-Tuning (SFT) for debate optimization, whereas we demonstrate that RL-based alternatives achieve superior performance.

3 Formalizing Self-Consistency

Given a prompt x, an LM with parameters θ defines a distribution $\pi_\theta(y \mid x) = \prod_{t=1}^{|y|} \pi_\theta(y_t \mid x, y_{<t})$ over reasoning trajectories y, from which answers $a = \mathrm{A}(y)$ are extracted. Under temperature sampling, the model samples from a

modified distribution $\pi_{\theta,\tau}(y \mid x)$ where token probabilities are adjusted by temperature $\tau > 0$. This induces an answer distribution $P_{\theta,\tau}(a \mid x) = \sum_{y:\mathrm{A}(y)=a} \pi_{\theta,\tau}(y \mid x)$, which gives each answer's probability by summing over all reasoning paths that lead to it. We denote the majority answer as $a^\star_{\theta,\tau}(x) = \arg\max_a P_{\theta,\tau}(a \mid x)$ with majority probability $S^+_{\theta,\tau}(x) = P_{\theta,\tau}(a^\star_{\theta,\tau}(x) \mid x)$. This represents the total probability mass concentrated on the most likely answer, or the model's internal consensus.

Temperature sampling enables exploration of diverse reasoning paths, but reduces the consistency of the final answer. While greedy decoding ($\tau = 0$) trivially approaches perfect consistency, it eliminates exploration and often produces suboptimal solutions (Holtzman et al., 2020). Lower temperatures increase consistency but restrict reasoning diversity. A self-consistent model should maintain high $S^+_{\theta,\tau}(x)$ even at high temperatures, allowing the model to access diverse reasoning trajectories while reliably converging on consistent answers. We measure self-consistency in two ways.

Single-agent sampling consistency. Computing $S^+_{\theta,\tau}(x)$ directly requires summing probabilities over all trajectories that lead to the majority answer, which is untractable. Instead, we estimate it by sampling t independent trajectories with answers $a_1, \ldots, a_t$ and computing:

$$s_t^{\theta,\tau}(x) = \frac{1}{t} \sum_{i=1}^{t} \mathbf{1}[a_i(x) = \hat{a}(x)], \quad \text{where } \hat{a}(x) = \text{Majority}\{a_1(x), \ldots, a_t(x)\}. \tag{1}$$

This measures the fraction of sampled trajectories that agree with the majority answer. As $t \to \infty$, $s_t^{\theta,\tau}(x) \to S^+_{\theta,\tau}(x)$, providing a consistent estimate of the true majority probability.

Multi-agent debate agreement. When M agents produce answers $a_1(x), \ldots, a_M(x)$ through deliberation, we measure the fraction of agents converging on the majority answer:

$$d_M^{\theta,\tau}(x) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}[a_m(x) = \hat{a}(x)], \quad \text{where } \hat{a}(x) = \text{Majority}\{a_1(x), \ldots, a_M(x)\}. \tag{2}$$

Higher agreement indicates a stronger consensus. In other words, models with higher $S^+_{\theta,\tau}(x)$ reach the same conclusion more frequently.

4 MACA: Multi-Agent Consensus Alignment

Having formalized self-consistency, we now present a framework to improve it through post-training with self-generated signals from debate. In multi-agent debate, M copies of the same model engage in iterative discussion: each agent generates an initial response, then all agents see each other's reasoning and update their answers over $R - 1$ subsequent rounds of deliberation. Answers that persist indicate stronger reasoning. The framework requires no external supervision: agents supervise themselves by learning from their own debate dynamics. Specifically, for each prompt x, the debate produces final responses $\mathcal{Y}(x) = \{y_1, \ldots, y_M\}$ with extracted answers $a_m = \mathcal{A}(y_m)$. The majority consensus $\hat{a}(x) = \text{Majority}\{a_1, \ldots, a_M\}$ partitions $\mathcal{Y}(x)$ into consensus-supporting $\mathcal{G}^+(x) = \{y \in \mathcal{Y}(x) : \mathcal{A}(y) = \hat{a}(x)\}$ and dissenting $\mathcal{G}^-(x) = \{y \in \mathcal{Y}(x) : \mathcal{A}(y) \neq \hat{a}(x)\}$ trajectories. This creates a fixed post-training dataset $\mathcal{D}_{\text{post}} = \{(x, \hat{a}(x), \mathcal{G}^+(x), \mathcal{G}^-(x))\}_{x \in \mathcal{D}}$ where $\mathcal{D}$ is the original set of prompts. Debate consensus, arising through deliberative exchange rather than statistical sampling, provides rich training signals. We adapt four post-training objectives to this self-generated data, treating consensus-supporting trajectories ($\mathcal{G}^+$) as preferred and dissenting trajectories ($\mathcal{G}^-$) as not preferred. By learning to separate these groups, the model internalizes the nuanced differences between stable consensus and dissenting reasoning. See Alg. 1 in App. C for the complete iterative debate and post-training loop.

Majority-Vote SFT (MV-SFT) trains the model to mimic consensus-supporting trajectories:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y^+ \in \mathcal{G}^+(x)} [\log \pi_\theta(y^+ | x)]. \tag{3}$$

Majority-Vote GRPO (MV-GRPO) uses online sampling with consensus-based rewards. For each prompt x, we sample multiple trajectories from the current policy and assign reward $r_x(y) = \mathbf{1}[\mathcal{A}(y) = \hat{a}(x)]$ based on whether each